(2) ${}_{80}Hg^{197} \rightarrow {}_{79}Au^{197} + {}_{1}e^{0}$

The mercury nucleus ${}_{80}Hg^{196}$ has changed into the gold nucleus ${}_{79}Au^{197}$.

Thus, fundamentally, the nuclear physicist would have no difficulty in producing gold out of mercury. However, this transmutation has never yet been actually recorded.

One might wonder why this transmutation has never been performed as yet. But the reason is that the profit would be far too small. Unfortunately, the mercury isotope ${}_{80}Hg^{196}$ is extremely rare; it represents not more than about 0·1 per cent. of the natural isotopic mixture of mercury. If mercury is bombarded by neutrons, only one neutron in a thousand will happen to become incorporated in one of these nuclei. From the less rare isotopes we cannot produce gold, but either another mercury isotope or thallium. It is, of course, conceivable that our goal could be reached more easily by a bombardment by fast neutrons. The mercury isotope ${}_{80}Hg^{198}$ is about 100 times more frequent than ${}_{80}Hg^{196}$. If we succeeded in heating this mercury nucleus of mass number 198, by neutron bombardment, to such a high degree that it would emit two neutrons simultaneously, we would get the gold nucleus ${}_{79}Au^{197}$, viz.:

(1) ${}_{80}Hg^{198} + {}_{0}n^{1} \rightarrow {}_{80}Hg^{197} + {}_{0}n^{1} + {}_{0}n^{1}$

(2) ${}_{80}Hg^{197} \rightarrow {}_{79}Au^{197} + {}_{1}e^{0}$

But in order to nip false hopes in the bud, let it be stated here that this method would indubitably be many million times more costly than the customary methods of getting gold.